STANES SCHOOL

1044, Avinashi Road, Coimbatore – 18.

UNIT TEST – 2 2026-2027

Class : XII **Duration : 2 Hours**

Subject : Computer Science (083) **Max. Marks : 50**

Section A (14 x 1 = 14 Marks)

1. State if the following statement is True or False:

The writelines() method in Python can be used to write a list of strings to a text file, and it automatically adds a newline character after each string.

2. Which of the following modes will open a file for both reading and writing in binary format, creating a new file if it doesn't exist?

a) rb+ b) wb+ c) ab+ d) rb

3. In the context of a stack, the operation to add an element is known as:

a) Push b) Pop c) Insert d) Append

4. In a relational database, a column of a table is referred to as a/an:

a) Tuple b) Attribute c) Domain d) Cardinality

5. Which method is used to read a single line from a text file in Python?

a) read() b) readline() c) readlines() d) scan()

6. In SQL, which constraint ensures that all values in a column are different?

a) NOT NULL b) UNIQUE c) PRIMARY KEY d) Both b) and c)

7. What will be the output of the following Python code?

```
import csv
with open('data.csv', 'w', newline='') as f:
    writer = csv.writer(f)
    writer.writerow(['Name', 'Age'])
    writer.writerow(['Alice', 25])
```

a) The code will create a CSV file named 'data.csv' with two rows.

b) The code will create a CSV file named 'data.csv' with one row.

c) The code will raise an error because the file is opened in write mode.

d) The code will append data to an existing 'data.csv' file.

8. Which SQL keyword is used to retrieve only unique values from a column?

a) UNIQUE b) DISTINCT c) DIFFERENT d) ALTER

9. What is the most appropriate module to use for performing operations on a binary file in Python?

a) csv b) os c) pickle d) sys

10. Which of the following is the correct syntax to display the structure of a table named 'STUDENT' in MySQL?

a) SHOW TABLE STUDENT; b) DESCRIBE STUDENT;

c) SELECT * FROM STUDENT; d) DISPLAY STUDENT;

11. Consider the following code:

```
f = open("sample.txt", "w")
f.write("Hello World")
f.close()
```

What is the mode of the file sample.txt after this code is executed?

a) Read mode b) Write mode

c) Append mode d) Read and Write mode

12. What is the degree of a relation that has 4 attributes and 10 tuples?

a) 4 b) 10 c) 14 d) 40

13. A stack is implemented using a list stack = []. Which of the following correctly implements the push operation?

a) stack.push(item) b) stack.append(item)

c) stack.insert(0, item) d) stack.add(item)

14. In SQL, which clause is used to filter records based on a condition after grouping?

a) WHERE b) GROUP BY c) HAVING d) ORDER BY

Section B (4 x 2 = 8 Marks)

15. A. Differentiate between the read() and readlines() methods for reading data from a text file.

OR

B. Explain the concept of a primary key and a foreign key in a relational database with a suitable example.

16. Write the output of the following Python code:

```
with open("story.txt", "w") as f:
    f.write("The quick brown fox jumps over the lazy dog.")
with open("story.txt", "r") as f:
    f.seek(10)
    data = f.read(9)
    print(data)
```

17. A. Write a Python statement to import the pickle module and then a statement to read an object from a binary file named data.dat.

OR

B. Write a SQL query to create a table named EMPLOYEE with the following attributes:

- Emp_ID (Integer, Primary Key)
- Emp_Name (String of size 50, Not Null)
- Salary (Float)

18. Consider the following implementation of a stack using a list:

stack = [10, 20, 30]

Write the Python code for the pop operation, which removes and returns the top element from the stack, and displays "Stack Underflow" if the stack is empty.

Section C (2 x 3 = 6 Marks)

19. A. Write a Python function named count_the() that reads the contents of a text file "ARTICLE.TXT" and displays the number of times the word "the" (case-insensitive) appears in the file.

OR

B. Write a SQL query for the following based on a table PRODUCT with columns P_ID, P_Name, Price, Qty:

I. To display the product names and their prices for products whose quantity is between 10 and 50.

II. To display the average price of all products.

III. To display the maximum price for each product category (Assume a column named Category exists).

20. A. A list of employees is given as:

employees = [("E1", "Alice", 60000), ("E2", "Bob", 45000), ("E3", "Charlie", 70000), ("E4", "Diana", 50000)]

Write the following user-defined functions to perform operations on a stack named Emp_Stack:

I. Push_emp() - To push the records of employees whose salary is greater than 50000 onto the stack.

II. Pop_emp() - To pop the items from the stack and display them. Also, display 'Stack is Empty' when there are no elements in the stack.

OR

B. Write a Python function add_records() that accepts a list of lists data (where each inner list is a record with fields [Name, Age, City]) and writes it to a CSV file named records.csv.

Section D (3 x 4 = 12 Marks)

21. A. Consider a CSV file "SALES.csv" with the following columns: Product_ID, Product_Name, Quantity, Price. Write Python functions to:

I. append_record() - Accept the Product_ID, Product_Name, Quantity, and Price from the user and add the record to the end of the "SALES.csv" file.

II. calculate_total_revenue() - Calculate and return the total revenue generated from all sales (revenue = Quantity * Price).

OR

B. Write a Python program that reads a text file "DATA.TXT" and creates a new file "EVEN.TXT" containing only the lines from "DATA.TXT" that have an even number of characters. Also, count and display the total number of lines written to "EVEN.TXT".

22. Consider two tables BOOKS and AUTHORS given below:

Table: BOOKS

Book_ID	Title	Author_ID	Genre	Price
B101	The Alchemist	A01	Fiction	450
B102	Sapiens	A02	Non-Fiction	600
B103	Harry Potter	A03	Fiction	800
B104	Wings of Fire	A04	Autobiography	350
B105	1984	A05	Fiction	500
B106	The Hobbit	A03	Fiction	700
B107	A Brief History of Time	A06	Science	950

Table: AUTHORS

Author_ID	Author_Name	Country
A01	Paulo Coelho	Brazil
A02	Yuval Noah Harari	Israel
A03	J.K. Rowling	UK
A04	A.P.J. Abdul Kalam	India
A05	George Orwell	UK
A06	Stephen Hawking	UK

Write SQL queries for the following:

I. To display the Title, Author_Name, and Price of all Fiction books.

II. To display the Country and the total number of books written by authors from each country.

III. To display the Title and Author_Name of books that have a price greater than the average price of all books.

IV. To display the Title, Author_Name, and Price of books sorted by Author_Name in ascending order and Price in descending order.

23. Consider two tables STUDENTS and EXAMS given below:

Table: STUDENTS

Roll_No	Student_Name	Class	City
101	Rajesh Kumar	12	Delhi
102	Priya Sharma	12	Mumbai
103	Amit Singh	12	Delhi
104	Sneha Reddy	11	Chennai
105	Vikram Joshi	12	Mumbai

Table: EXAMS

Roll_No	Subject	Score	Exam_Date
101	Mathematics	95	2025-01-15
101	Science	88	2025-01-16
102	Mathematics	92	2025-01-15
103	Mathematics	78	2025-01-15
104	Science	85	2025-01-16
105	Mathematics	91	2025-01-15
102	Science	94	2025-01-16

Write SQL queries for the following:

I. To display the Cartesian Product of the STUDENTS and EXAMS tables.

II. To display the Student_Name, Subject, and Score of all students using an equi-join.

III. To display the Student_Name, Subject, and Score of students who have scored more than 90 in any subject.

IV. To display the Student_Name and the average score for each student, but only for those students whose average score is greater than 85.

Section E (2 x 5 = 10 Marks)

24. Vidhi, a data analyst, is working on a large text file "LOG.txt" that contains log entries. Each line in the file has a timestamp and a message. Help her create the following user-defined functions:

I. count_error_lines() - to count and return the number of lines in the "LOG.txt" file that contain the word "ERROR".

II. create_error_report() - to read the "LOG.txt" file and write all lines that contain the word "WARNING" to a new file "WARNINGS.txt".

III. find_keyword_position() - to take a keyword as an argument and find the first occurrence of the keyword in the "LOG.txt" file using the seek() and tell() methods, printing the position of the keyword if it is found.

25. A. Mr. Anuj, the manager of a bookstore, needs to maintain a stack of books in the store's inventory. Each record includes: Book_ID, Book_Title, Author, Price, Stock.

Write the Python functions to:

I. push_book() - Input book data from the user and push it onto a stack (implemented using a list).

II. pop_book() - Pop the top book record from the stack and display it.

III. display_stack() - Display all the books present in the stack without removing them.

OR

B. Consider the following table **EMPLOYEE**:

Emp_ID	Emp_Name	Department	Salary	Joining_Date
E101	Raj Sharma	IT	75000	2020-05-15
E102	Priya Singh	HR	55000	2019-08-20
E103	Amit Kumar	Finance	65000	2021-01-10
E104	Sneha Reddy	IT	80000	2018-11-25
E105	Vikram Joshi	Marketing	60000	2022-03-01
E106	Sunil Verma	IT	90000	2020-07-12
E107	Ananya Shah	HR	48000	2021-09-05

Predict the output of the following SQL queries:

I. SELECT Emp_Name, Salary FROM EMPLOYEE WHERE Department = 'IT';

II. SELECT Department, AVG(Salary) FROM EMPLOYEE GROUP BY Department;

III. SELECT * FROM EMPLOYEE WHERE Emp_Name LIKE 'S%';

IV. SELECT COUNT(*) FROM EMPLOYEE WHERE Joining_Date > '2020-01-01';

V. SELECT Department, COUNT(*) AS Total_Employees, SUM(Salary) AS Total_Salary FROM EMPLOYEE GROUP BY Department;